Performance Analysis of a Hankel-Structured Channel Estimator for Rydberg Atomic MIMO Receivers With Magnitude-Only Observations

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Abstract*—Multi-user channel estimation for an atomic receiver measuring only the magnitude of the incident field, in the presence of a known additive reference beam, is studied in this letter. The problem is one of biased phase retrieval, for which the Fisher information is shown to be rank one per measurement and block diagonal across the receive elements, so that the normalised per-element information is independent of the array size and the aperture is exploitable only through cross-element structure. For a uniform linear array, the Hankel lifting of each channel column supplies such a structure with a capacity of $\lceil N/2 \rceil$ ranks, from which it is predicted that a rank prior is informative only when the channel occupies a small fraction of that capacity, and harmful otherwise. An estimator enforcing this structure within an exact expectation–maximization Gerchberg–Saxton iteration, with the model order selected from a held-out pilot residual, is proposed. The predicted behaviour is observed: a gain of $+2.93$ dB at $N = 32$ and a loss of 0.25 dB at $N = 8$ in summed squared error, averaged over twelve operating points per aperture, the unstructured baseline varying by at most 0.163 dB. The reversal is a property of that statistic alone; under a paired per-trial median the same $N = 8$ cells yield $+0.225$ dB, the selector abstaining in 42% of the trials. The governing variable is shown to be the effective rank relative to capacity rather than the path count, and predicts the gain on two channel distributions to which it was not fitted, with errors of 0.13 and 0.315 dB. Results are simulation only, and the derived bounds do not govern biased estimators.

***Index Terms*—Channel estimation, constrained Cramér–Rao bound, effective rank, Hankel matrix, phase retrieval, Rydberg atomic receiver.**

I. Introduction

ATOMIC receivers based on Rydberg states have emerged as a promising technology for radio-frequency sensing and reception, since they read out the incident field through an optical transition whose response is proportional to the field *magnitude*, thereby discarding the phase [1], [2]. Owing to this, multi-user channel estimation in such receivers is a biased phase-retrieval problem, for which Gerchberg–Saxton-type alternating projection [3] is the natural solver. The atomic receiver enters only through the observation model (1), a magnitude-only measurement with a known additive reference; nothing below depends on the vapour-cell physics, so the results apply to any array receiver with that measurement.

Several studies on channel estimation for such receivers have been reported. In [4], the Gerchberg–Saxton iteration is unrolled into a trained network. In [5], a rank projection is placed inside projected gradient descent for the same problem, and the geometric channel model and path-count distribution used here are specified. Structured low-rank Hankel recovery for sums of exponentials has been studied outside this setting: EMaC [6], ALOHA [7] and LORAKS [8] recover such a matrix from *partial linear* observations by convex relaxation, while gridless line-spectrum and atomic-norm methods [9] likewise assume linear measurements and a separation condition. Both families differ from the present setting in the same way: the measurement here is a *modulus*, so the map is not linear, no convex relaxation applies, and the projection is a regulariser inside a non-convex iteration rather than the estimator itself. No new projection algorithm is claimed; Cadzow's composite property mapping [10] is used as given.

In this letter, we consider the conditions under which a structural prior is informative for a magnitude-only estimator, organising the work around a single argument stated first and tested afterwards. Under the unconstrained row-wise parameterisation of (1) the Fisher information is block diagonal across the receive elements, so the normalised per-element information does not grow with N and the aperture is usable only through cross-element structure. On a uniform linear array (ULA) each path contributes one spatial complex exponential, so the Hankel lifting of every channel column is rank deficient [10], [11] and admits at most $r_{\max}(N) = \lceil N/2 \rceil$ ranks, whence a rank prior is informative only when the channel occupies a small fraction of that capacity. At $N = 8$ the capacity is 4 while the path counts are $L_k \sim \mathcal{U}\{3, \ldots, 7\}$, so the constraint is frequently vacuous and, when active, truncates genuine components. A sign reversal at small N is thus predicted rather than discovered afterwards, and is reported as such in Section IV-B in summed squared error, the quantity a bound governs; it does not appear in a median, for a reason made precise in Section IV-D.

The remaining difficulty lies in determining when the channel does occupy a small fraction of the capacity. An appeal to sparsity is unusable, since a 16-path channel on a 32-element array is algebraically full rank yet has an effective rank of 8.73, its lifting being compressible with nothing sparse about it. The path count is accordingly replaced by the effective rank relative to capacity, which is then tested prospectively. The major contributions of this letter can be summarized as follows.

- The Fisher information is derived and shown to be rank one per measurement and block diagonal across the

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receive elements, and a geometry-constrained bound is obtained in tangent-space form, the textbook form being inapplicable owing to rank deficiency of the parameterisation.
- The capacity $r_{\max}(N) = \lceil N/2 \rceil$ is derived and the aperture-scaling behaviour is presented as a test of a prediction made in advance of the measurement.
- The path-count heuristic is replaced by $\rho = r_{\text{eff}}/r_{\max}$, with a measurement separating the contribution of the *prior* from that of the *estimator*.
- Two cross-generator predictions registered before measurement are reported, together with the pre-registered criteria that were not met.

II. System Model and Proposed Estimator

Consider an N-element ULA serving K single-antenna users over P pilots. With channel matrix $\mathbf{G} \in \mathbb{C}^{N\times K}$, pilots $\mathbf{S}$, known reference beam $\mathbf{B}$ and noise $\mathbf{W}$, the received signal is given by

$$\mathbf{Z} = |\mathbf{GS} + \mathbf{B} + \mathbf{W}|, \qquad \mathbf{W}_{np} \sim \mathcal{CN}(0, \sigma^2), \tag{1}$$

where $|\cdot|$ denotes the elementwise magnitude operator. The modulus is applied after the noise, owing to which (1) cannot be linearised. The channel of each user is a sum of L_k plane waves on the array manifold,

$$g_k[n] = \sum_{\ell=1}^{L_k} \alpha_{\ell,k} e^{-j(n-1)\psi_{\ell,k}}, \qquad \psi_{\ell,k} = \pi \sin\theta_{\ell,k}, \tag{2}$$

where $j = \sqrt{-1}$, $L_k \sim \mathcal{U}\{3, 7\}$ [5] and $\theta \sim \mathcal{U}(-90°, 90°)$. The reference-to-signal ratio (RSR) is the reference power divided by the *single-user* received power.

A. Structure and Capacity of the Hankel Lifting

Substituting (2) into the $(N-p) \times (p+1)$ Hankel lifting $\mathcal{H}_p$ separates the exponential in the two indices, so each path contributes one rank-one outer product and $\operatorname{rank} \mathcal{H}_p(\mathbf{g}_k) \le L_k$. The maximum admissible rank is therefore

$$r_{\max}(N) = \max_p \min(N-p,\, p+1) = \lceil N/2 \rceil, \tag{3}$$

and the constraint is vacuous whenever $L_k \ge r_{\max}(N)$. Note that K cancels: the structural model employs $3\sum_k L_k$ parameters against $2NK$, a ratio independent of K, so the gain is expected to be K-invariant at fixed pilot adequacy, as tested in Section IV-F.

B. The Proposed Estimator

The proposed Hankel-structured Gerchberg–Saxton (HS-GS) estimator applies, after every exact expectation–maximization Gerchberg–Saxton (EM-GS) update, a Cadzow projection $\mathcal{P}_r(\mathbf{g}) = \mathcal{H}_p^\dagger(\Pi_r(\mathcal{H}_p(\mathbf{g})))$, where Π_r truncates to the r leading singular values and $\mathcal{H}_p^\dagger$ averages the antidiagonals. A single application is *not* idempotent, being one step of an alternating projection between the rank set and the Hankel set, so the sweep count forms part of the definition of the operator. The settings are not uniform across Section IV: every result below is obtained at one of the four configurations of Table I and is tagged accordingly. All four employ $n_{\text{cz}} = 4$ sweeps and differ in T and the selection budget. With the projection disabled, HS-GS reproduces EM-GS to $\max|\text{diff}| = 0$, so the projection is provably the only difference.

C. Model Order Selection Without an Oracle

A fraction $\nu = 0.3$ of the pilot columns is held out, each candidate order $r \in \{1, \dots, r_{\max}\}$ is scored by the held-out magnitude residual, and the minimiser is retained. The in-sample residual cannot serve, since constraining $\mathbf{G}$ can only worsen the fit to the pilots used for fitting, so an in-sample criterion always prefers the largest candidate. The true L_k never enters, so the rule uses no oracle. When the selector returns $\hat{r} \ge r_{\max}$ the projection reduces to the identity and the estimator *abstains*, falling back exactly to EM-GS, a designed behaviour shown in Section IV-D to be load-bearing.

Subspace methods such as root-MUSIC and ESPRIT recover $\{\psi_{\ell,k}\}$ from the signal subspace, but that subspace requires either multiple snapshots or the complex field. With magnitude-only single-snapshot data it is identifiable only *after* phase retrieval has succeeded, at which point the channel is already estimated. The exponential-sum manifold is therefore regularised toward rather than parameterised.

Throughout, the primary statistic is the paired per-trial median within an SNR bin, with bootstrap 95% confidence intervals. Where a curve is compared against a bound the ratio of summed errors is used, this being what a bound on the mean-squared error governs; both are reported for the headline cell.

III. Performance Benchmark

A. Rank-One Fisher Information

Conditioned on $\mathbf{G}$, $\mathbf{S}$ and $\mathbf{B}$ the noiseless field at (n, p) is a constant λ, so $z = |\lambda + w|$ is Rice distributed with a density depending on λ *only through* $a = |\lambda|$. Denoting $\frac{d}{dx}\log I_0(x) = R(x)$ and $\kappa = 2za/\sigma^2$, the score is $\partial_a \log p = \frac{2}{\sigma^2}[zR(\kappa) - a]$. Writing $\beta = \sigma^{-4}(\mathbb{E}[z^2R^2(\kappa)] - a^2)$, one measurement carries the scalar information 4β, so in the real coordinates $\boldsymbol{\eta} = [\Re\text{vec}\,\mathbf{G}; \Im\text{vec}\,\mathbf{G}]$,

$$\mathbf{J}_n = \sum_{p=1}^{P} 4\beta(a_{n,p}, \sigma^2)\, \mathbf{u}_{n,p}\mathbf{u}_{n,p}^T. \tag{4}$$

Each magnitude measurement thus contributes rank-one and not rank-two information. Crediting both quadratures overstates it and yields a bound too low by 0.17–4.64 dB across the 36 operating points considered. An identifiability check settles the form: at $K = 3$, $P = 3$ the per-row problem is unidentifiable and (4) is correctly singular, whereas the rank-two construction claims full rank. Since the n-th row of $\mathbf{GS}$ enters only the n-th measurement row, $\mathbf{J}$ is *block diagonal across the receive elements*, so an unstructured estimator — indeed any estimator attaining the unconstrained bound — is flat in N under a normalised metric.

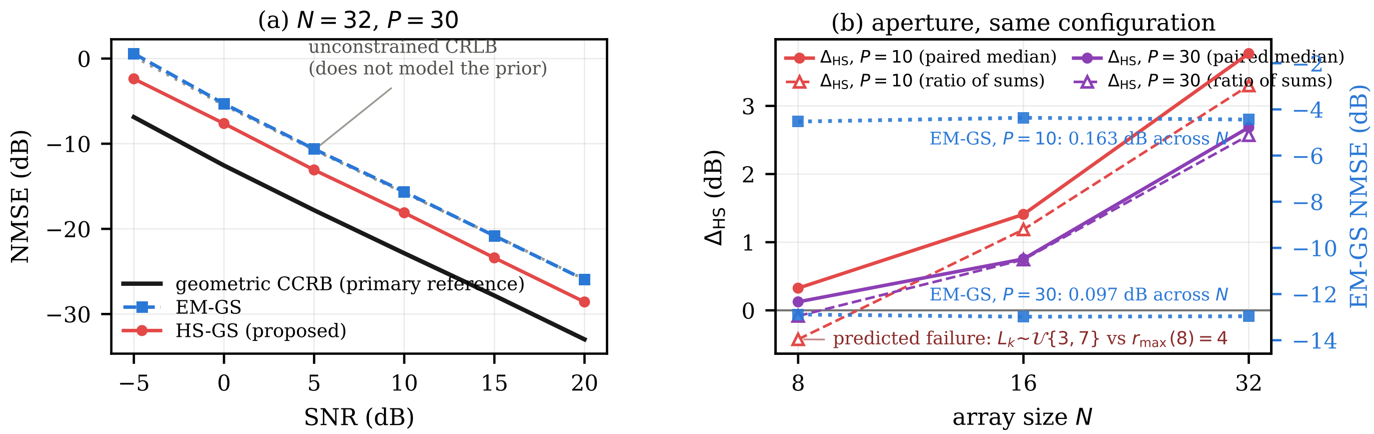

Fig. 1. Configuration A ($n_{\mathrm{cz}} = 4$, $T = 100$, budget 25, RSR 12 dB, $K = 3$, 400 paired trials/point), both panels sharing one operator. (a) NMSE versus SNR at $N = 32$, $P = 30$; the unconstrained bound is faint since it does not model the prior, and neither bound governs a biased estimator. (b) Δ_{HS} versus N, each point the mean over six SNR values; filled markers are the paired median, open markers the ratio of summed errors, the two disagreeing at $N = 8$.

B. Constrained Bound on the Geometric Manifold

The unconstrained bound governs estimators treating $\mathbf{G}$ as $2NK$ free parameters. The proposed estimator instead exploits (2), whose parameters $\boldsymbol{\phi} \in \mathbb{R}^{3\sum_k L_k}$ number about 45, *independent of* N. The textbook form $\mathbf{D}(\mathbf{D}^T\mathbf{J}\mathbf{D})^{-1}\mathbf{D}^T$ with $\mathbf{D} = \partial\boldsymbol{\eta}/\partial\boldsymbol{\phi}$ is unusable, $\mathbf{D}$ being rank deficient: at $N = 8$ its mean numerical rank is 40.4 against ≈ 44.7 parameters. The tangent-space form of [13], [14] is therefore employed, with $\mathbf{U}$ an orthonormal basis of $\mathrm{range}(\mathbf{D})$,

$$\mathrm{CCRB} = \mathbf{U}(\mathbf{U}^T\mathbf{J}\mathbf{U})^{-1}\mathbf{U}^T, \tag{5}$$

which depends on the tangent *space* and is invariant to over-parameterisation. Both bounds are averaged over the same realisations as the estimator curves.

Both bounds constrain the covariance of *unbiased* estimators. All estimators here run to a fixed budget with a shrinking update, and HS-GS also truncates the rank, so none is unbiased and neither bound applies strictly. The constrained bound is the primary reference in Fig. 1, being matched to the estimator's model class; the unconstrained bound is secondary and governs EM-GS only.

IV. Numerical Results

This section provides the numerical validation of the results derived above. For the simulations, both estimators are handed identical realisations and $K = 3$, except where K is the swept variable. Everything else varies by configuration, so Table I states the four in use and every number below carries its label. Configurations A and B differ only in T and the selection budget, the one cell measured both ways differing by 0.005 dB.

A. Performance Against the Derived Bounds

In Fig. 1(a), the NMSE of both estimators is plotted against SNR for configuration A at $N = 32$, $P = 30$, together with the two bounds of Section III. HS-GS improves upon EM-GS by $+2.46$ to $+3.05$ dB across the sweep, with a six-point mean of $+2.68$ dB and a ratio-of-sums improvement of $+2.33$ to $+2.96$ dB, the projection being active in 100% of the trials at this cell.

EM-GS tracks the unconstrained bound to within 0.143 dB for SNR ≥ 5 dB, so it is informative rather than vacuous. The constrained bound lies 7.05–7.11 dB *below* it here, that gap being what the structural model is worth in principle; HS-GS recovers part of it, remaining above the constrained bound at all six points by 4.39–4.93 dB.

TABLE I

Configurations used in Section IV: same estimator, same held-out order rule ($\nu = 0.3$, candidates $1..r_{\max}$, no oracle) and $n_{\mathrm{cz}} = 4$ sweeps throughout, differing only in operating point. TB = Track B, TD = Track D generator; "$\mathcal{U}$" denotes SNR drawn per trial and binned post hoc. †P is swept in the K-invariance (13–27) and pilot-count (10–35) cells.

	T	sel.	P	RSR	Gen.	trials	SNR (dB)	used in
A	100	25	10, 30	12	TB	300–400	$\{-5..20\}$	§IV-A, §IV-B, §IV-F
B	50	20	30	12	TB	400	$\{-5..20\}$	§IV-F (timing)
C	100	25	20†	10	TD	60–1000	$\mathcal{U}[-10, 20]$	§IV-D–§IV-F
D	50	20	30	12	TB	300	5	§IV-C, §IV-D

B. Aperture Scaling and the Choice of Statistic

Section III predicts that the gain turns negative at small aperture. In Fig. 1(b), Δ_{HS} is plotted against $N \in \{8, 16, 32\}$ at configuration A, each point being the mean over 12 operating points (2 pilot counts $\times$ 6 SNR values), so this panel and Fig. 1(a) share one operator. The two pooling rules disagree at $N = 8$ and agree elsewhere, so both are reported below.

N	ratio of sums	paired median	EM-GS NMSE
8	-0.254	$+0.225$	-8.705
16	$+0.966$	$+1.081$	-8.672
32	$+2.934$	$+3.226$	-8.700

It is evident that EM-GS varies by at most 0.163 dB over the same fourfold change in aperture, namely 0.163 dB at $P = 10$ and 0.097 dB at $P = 30$, against a structured gain moving by over 3 dB. Pooling the pilot counts first yields 0.033 dB, but the two move in opposite directions, so the larger figure is quoted. This is what Section III requires of an estimator whose information is block diagonal in N.

The predicted reversal is a ratio-of-sums phenomenon. At $N = 8$ the projection is inactive in 41.8% of the trials on average, rising from 12% at -5 dB to 59% ($P = 10$) and 79% ($P = 30$) at $+20$ dB as the selector declines a constraint it cannot afford. Where active it truncates genuine components, and the ratio of summed errors turns over accordingly, from $+1.272$ dB at -5 dB to -2.613 dB at $+20$ dB for $P = 10$, truncation bias not shrinking with the noise while the variance term does. This is the predicted failure, measured.

The paired median does not reverse, and no claim is made that it does. At $N = 8$ it is $+0.000$ dB at nine of the twelve points, positive and significant at the other three, and negative and significant at none. The mechanism is that of Section IV-D: abstention renders most trials bit-identical to EM-GS and pins the median at zero, while the active minority dominates a sum of squared errors. The prediction is therefore confirmed on the statistic a bound governs and is *invisible* to the one otherwise treated as primary. A reader taking only the median away would conclude that the prior is never harmful, and Section IV-D shows that it is not.

The bands registered before this sweep was run were $[-0.35, +0.05]$, $[+0.60, +0.95]$ and $[+2.65, +3.00]$ dB. The values at $N = 8$ and $N = 32$ lie inside their bands and that at $N = 16$ misses by 0.016 dB, so the pre-registration failed. The EM-GS spread was predicted to be at most 0.15 dB; it is 0.033 dB pooled, which the prediction named, and 0.163 dB at $P = 10$ alone, which would fail it. A failed prediction licenses no reporting other than the measurement.

C. Path Count as a Coordinate

At configuration D, holding $N = 32$ and varying only the path count, with L identical across users at 5 dB SNR, the gain decays monotonically from $+7.04$ dB at $L = 2$ to -0.12 dB at $L = 16 = r_{\max}$, while EM-GS moves by 0.191 dB in total. The operative variable is therefore channel complexity relative to capacity, and aperture matters only because it sets that capacity.

D. Effective Rank, and the Prior Versus the Estimator

The path count is nevertheless imperfect, since clustered or unequal-strength paths occupy fewer effective dimensions. The results are therefore indexed by the Roy–Vetterli effective rank [12] of the lifting, $r_{\text{eff}} = \exp(-\sum_i p_i \ln p_i)$ with $p_i = \sigma_i / \sum_j \sigma_j$, normalised as $\rho = r_{\text{eff}}/r_{\max}$.

Re-indexing the sweep of Section IV-C (configuration D) moves the zero crossing from $L/r_{\max} = 0.90$ to $\rho = 0.518$, and the useful region becomes $\rho \lesssim 0.5$, a statement about any channel rather than about the L values of one simulator. In Fig. 2, Δ_{HS} is plotted against ρ for $N \in \{16, 32, 64\}$, the crossings being 0.588, 0.518 and 0.544, a spread of 0.070 over a fourfold change in aperture. This one comparison spans two configurations, the $N = 32$ value being at configuration D and the other two at configuration C, and only the $N = 32$ crossing is bracketed on both sides, the other two being extrapolated from the positive side for the reason given below.

The crossing belongs to the prior rather than to the estimator, and the two must be stated separately. Past the crossing at configuration C the paired per-trial median never becomes negative: at $\rho = 0.608$ and 0.735 for $N = 16$ it is $+0.078$ and $+0.000$ dB, and at $\rho = 0.596$ and 0.669 for $N = 64$ it is $+0.044$ and $+0.018$ dB. The reason is abstention, since at $\rho = 0.735$ the selector returns $\hat{r} \geq r_{\max}$ in 38.8% of the trials, the estimator then being exactly EM-GS. Forcing the projection active at that cell, as a diagnostic which is not the proposed method, yields -0.075 dB, CI $[-0.186, -0.026]$. The prior therefore does become harmful past the crossing, and abstention is what prevents the *typical* trial from following it down.

This does not contradict the negative values reported in Sections IV-B and IV-C; the difference is the statistic, by the mechanism given there. Consequently, $\Delta_{\text{HS}} \geq 0$ is a property of the selector under a median statistic only, and is neither evidence that the prior is never harmful nor a guarantee concerning the mean-square error; where the two disagree, both are reported.

The magnitude of the gain does not collapse onto the same axis. Over the window covered by both arrays at configuration C, $N = 16$ lies above $N = 64$ by a mean of 0.493 dB and a maximum of 0.901 dB, one-signed throughout, and no account of this residual is offered. The one candidate tested, that the rank grid of the selector is coarser at small capacity, is refuted: varying the pencil at fixed $N = 32$ and fixed ρ gives a span of 0.390 dB over admissible-rank counts of 8 to 16 with *no* monotone trend, the dependence running mildly backwards. A pair with opposite matrix shapes but nearly equal rank counts, 24×9 and 8×25, gives $+3.199$ and $+3.121$ dB, so the effect tracks the rank count and not the shape.

Finally, ρ is a design-time and not a receiver-observable coordinate, since r_{eff} is computed on the noiseless channel. The abstention rate is observable and monotone in ρ at fixed aperture, but is confounded by capacity: at $N = 64$ a cell at $\rho = 0.501$ abstains in 0.0% of the trials while at $N = 16$ a cell at $\rho = 0.371$ abstains in 4.0%, so no single threshold separates the useful region. An observable proxy therefore exists within an aperture but not across apertures.

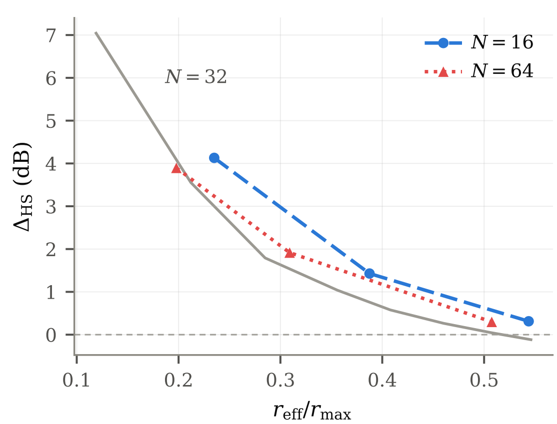

Fig. 2. Δ_{HS} versus $\rho = r_{\text{eff}}/r_{\max}$ for $N \in \{16, 32, 64\}$. The boundary location is stable across a fourfold change in aperture; the gain magnitude is not, and that residual is unexplained.

E. Cross-Generator Prediction

A relation fitted and evaluated on the same data proves little. The ρ-to-gain relation was therefore frozen, numerical predictions were committed to version control, and only then were two channel generators run which had played no part in constructing it. The rule is a piecewise-linear interpolation on a fixed eight-point table, with no regression and no spline, so it re-executes exactly. Both tests employ the operator and operating point of configuration C with the generator replaced.

For a clustered Saleh–Valenzuela generator at $\rho = 0.331$ the relation predicted $+1.30$ dB, and the measurement over 276 trials gave $+1.173$ dB, CI $[+1.017, +1.380]$, an error of 0.13 dB. For a second, independently specified configuration at $\rho = 0.400$ the prediction was $+0.648$ dB with interval $[+0.367, +1.015]$, and the measurement over 266 trials gave $+0.963$ dB, CI $[+0.808, +1.086]$, an error of 0.315 dB, inside the stated interval but on its upper edge.

These are *cross-generator* and not out-of-model tests, since both generators still produce channels of the form (2) and only the distribution over $(L_k, \alpha_{\ell,k}, \theta_{\ell,k})$ changes. They show that ρ transfers across path distributions, not that it survives a violation of the ULA model.

F. Users, Cost, and Placement of the Projection

At configuration C and fixed pilot adequacy $P/2K = 3.33$, the gain varies by 0.094 dB across $K \in \{2, 3, 4\}$ for SNR ≥ 5 dB, corroborating the K-invariance predicted in Section II. Pooled over all SNR the spread is 0.294 dB against a pre-registered falsifier of 0.3 dB, missed by 0.006 dB, and is reported rather than dropped.

Order selection accounts for 57.7–85.5% of the runtime. A coarse-to-fine stride-3 search at configuration C cuts that stage by $1.68\times$, from 16.0 to 9.6 candidate evaluations, and returns the same $\hat{r}$ in 93.3% of the trials, but missed its pre-registered criterion of at most 0.05 dB, the aggregate cost being 0.083 dB. Its paired median degradation of 0.000 dB understates that cost, for the reason abstention does above, so it is reported as an option which missed its gate and not as a validated substitute.

Comparing interleaved projection against a single post-hoc Cadzow projection at the same selected order, at configuration A, interleaving is *worse* at 0 dB (-0.164 dB, CI $[-0.317, -0.066]$), indistinguishable from $+5$ to $+15$ dB, and better only at $+20$ dB ($+0.168$ dB, CI $[+0.102, +0.246]$), the six-point mean being $+0.003$ dB. Interleaving is therefore not required in this classical iteration, and no claim is made that it is.

Since placement is a wash, what the estimator contributes is not *where* the projection is applied but *whether* it is applied at all and at what order, decided from held-out pilots and using no oracle. Section IV-D provides the evidence: the selector is what keeps $\Delta_{\mathrm{HS}} \geq 0$, and forcing the projection past the crossing costs -0.075 dB, CI $[-0.186, -0.026]$. The order selector is thus the claim of this letter and not a mechanism explained on the way to one.

V. Conclusion

Multi-user channel estimation for a magnitude-only atomic receiver with a known additive reference is considered in this letter. The Fisher information is shown to be rank one per measurement and block diagonal across the receive elements, so the aperture is usable only through cross-element structure. The Hankel lifting supplies such a structure with a capacity of $\lceil N/2 \rceil$ ranks, whence the prior helps only below a fraction of that capacity, and the predicted sign reversal at small aperture is observed. The governing coordinate is $r_{\mathrm{eff}}/r_{\max}$, whose boundary location is stable across a fourfold change in aperture and which predicted the gain on two unseen channel distributions to within 0.13 and 0.315 dB.

The scope is limited in the following respects. All results are obtained by simulation only, with no measured hardware and no recorded channel data. They cover geometric ULA channels of the form (2) at the array sizes, pilot counts and generators tested, and the cross-generator tests change the distribution but not the model family, so no claim is made beyond them. The magnitude of the gain does not collapse onto the proposed axis, its residual of 0.493 dB being unexplained, and ρ is not receiver-observable. Two of the three boundary crossings cannot be bracketed from the negative side, since abstention prevents it, and $\rho \approx 0.75$ is unreachable at $N = 64$, where the path generator fails above $L = 48$ ($\rho \approx 0.67$). No fundamental bound is established, since the derived bounds govern unbiased estimators while every estimator here is biased.

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